Fiscal Year:	FY 2005	**Task Last Updated:**	FY 10/21/2004
PI Name:	Fedorov, Andrei G Ph.D.		
Project Title:	MEMS-ENABLED PROCESSING OF LIQUID FUELS FOR DISTRIBUTED POWER GENERATION USING FUEL CELLS		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Droplets, sprays, particles, dust clouds, nanoparticles		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	30332	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Physical Sciences 02-OBPR-03
Start Date:	10/01/2003	**End Date:**	09/30/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	3	**No. of Master' Degrees:**	
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Degertekin, Levent (Georgia Institute of Technology)		
Grant/Contract No.:	NNC04GA12G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Fuel cell based portable power sources offer opportunities for high efficiency and ultra-high density distributed power generation, which is essential for enhanced capabilities and survivability of the future long duration space missions as well as for extra vehicular activities (EVA) of astronauts. A key element of the fuel cell power sources is the fuel processor, which should have a capability to produce hydrogen from a variety of hydrocarbon fuels stored in a liquid phase for highest power density. To this end, this project focuses on the novel portable MEMS hydrogen generator with improved functionality and low cost achieved through innovative system integration approach and by exploiting advantages pertinent to fluid mechanics, heat and mass transfer, and catalysis on micro/nano scale. The proposed microreactor has multifuel processing capability achieved by using unique piezoelectrically-driven fuel atomization technology, it operates in autothermal regime that does not require external heating, it exhibits high fuel conversion and		

Task Description:	selectivity to hydrogen, and it features low “skin” temperature which is essential from thermal management prospective. To establish the fundamental bases for optimal design and operation of the proposed catalytic microreactor, a ground-based three-year experimental study of the microreactor and its key components is undertaken. The key objectives include (1) design, fabrication, and testing of the autothermal MEMS catalytic reactor integrated with hydrogen separating membrane, (2) design, fabrication, and experimental characterization of the novel micromachined fuel ejector array for energy-efficient atomization and mixing of liquid fuels, and (3) integration and experimental characterization of an integrated microreformer that combines a microreactor, a hydrogen separating membrane, and an atomizer/droplet ejection array.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Summary of Accomplishments: 1. Scaling analysis and development of 1-D model for reverse-flow microreactor 2. Development of micromachined hydrogen separation membrane: 2.1 Design of fabrication process and initial fabrication, 2.2 Design and assembly of test fixture and complete test set-up, 2.3 Initial testing of membranes. 3. Development of micromachined ultrasonic fuel atomizer: 3.1 Compilation of fuel properties database, 3.2 Improved ANSYS simulation of atomizer frequency response and pressure fields, 3.3 Improved characterization of water ejection, 3.4 Successful ejection of methanol, kerosene, and butanol.
Bibliography Type:	Description: (Last Updated: 02/25/2014)
Abstracts for Journals and Proceedings	Kaisare N, Fedorov A, Lee JH. "Hydrogen generation in a microchannel reactor with periodic flow reversal: simulation and analysis." 2004 AIChE Annual Meeting, Austin, Texas, USA, November 7-12, 2004. AIChE 2004, Annual Meeting, November 2004. , Nov-2004
Abstracts for Journals and Proceedings	Meacham JM, Varady MJ, Esposito D, Degertekin FL, Fedorov AG. "A Micromachined Ultrasonic Atomizer for Liquid Fuels." Presented at ASM Symposium on Fuel Cells Materials, Processing, and Manufacturing Technologies, ASM International, Columbus, Ohio, USA, October 18-21, 2004. ASM Symposium on Fuel Cells Materials, Processing, and Manufacturing Technologies, ASM International, October, 2004. , Oct-2004
Papers from Meeting Proceedings	Meacham JM, Varady MJ, Degertekin FL, Fedorov AG. "Droplet formation and ejection from a micromachined ultrasonic droplet generator: visualization and scaling." Presented at the Conference on Transport Phenomena in Micro and Nano Devices, Engineering Conferences Foundation, Hawaii, USA, October 17-21, 2004. Conference on Transport Phenomena in Micro and Nano Devices, Engineering Conferences Foundation, October, 2004. , Oct-2004
Presentation	Fedorov, A. G. "Micromachined Ultrasonic Atomizer for Liquid Fuels" Invited Poster Presentation at the Workshop on Solid Oxide Fuel Cells, DOE Solid Energy Conversion Alliance (SECA), Boston, Massachusetts May-2004